

Writing and Solving Equations in Real-World Contexts*Additional Practice 7A.01.07*

Name _____

Date _____

Period _____

For questions 1 through 6, define the variable, write an equation, solve it, and say what the solution means.

1. Marcus buys five identical gift boxes online and pays a \$6.95 shipping charge on the order. The total came to \$91.70. What does one gift box cost?

Variable.

Equation.

Solve.

Meaning.

2. A bakery sold 12 fewer loaves today than twice the number it sold yesterday. Today it sold 5 times as many loaves as the café next door, which sold 8. How many loaves did the bakery sell yesterday?

Variable.

Equation.

Solve.

Meaning.

3. Jamal has a \$75 gift card. He buys six identical items and has \$13.50 left on the card. What did each item cost?

Variable.

Equation.

Solve.

Meaning.

4. A summer camp charges each of seven children a \$22 registration fee plus the same activity fee. The seven families paid \$364 in all. What is the activity fee?

Variable.

Equation.

Solve.

Meaning.

5. A granola recipe uses $\frac{2}{5}$ cup of oats for each serving, plus 1.4 cups of oats for the topping. The whole batch uses 7 cups of oats. How many servings does the recipe make?

Variable.

Equation.

Solve.

Meaning.

6. A gas bill charges \$0.98 for each therm used plus a \$16.50 service charge. Last month's bill was \$114.50. How many therms were used?

Variable.

Equation.

Solve.

Meaning.

Question 7 asks whether the answer makes sense. Question 8 is error analysis.

7. A field trip costs \$260 plus \$17 for each student. The budget is \$700. How many students can go?

Variable.

Equation.

Solve.

Meaning.

8. Rosa was solving $\frac{1}{8}s + 0.75 = 2$. She wrote $\frac{1}{8}s = 1.25$ and then $s = 1.25 \div 8 = 0.15625$. Explain her error and give the correct solution.

Questions 9 and 10 look ahead to multi-step equations, which arrive later this year. The unknown appears on both sides of the equal sign.

- 9.** Solve $4(n + 5) = 6n - 2$. Check your solution by substitution.

Equation.

Solve.

Meaning.

- 10.** One tank holds 240 gallons and is draining at 8 gallons a minute. A second tank holds 60 gallons and is filling at 12 gallons a minute. After how many minutes do the two tanks hold the same amount?

Variable.

Equation.

Solve.

Meaning.